

Implementation of an Anomaly Detection System for Gas Consumption in Industrial Clients through Data Analytics

María Alejandra Benavides Sanclemente
Industrial Engineering
Universidad de los Andes
Bogota, Colombia
ma.benavides1@uniandes.edu.co

María Camila Luna Velasco
Industrial and Systems Engineering
Universidad de los Andes
Bogota, Colombia
m.lunav@uniandes.edu.co

Abstract

The natural gas industry, plays a critical role worldwide in the current energy transition, serving as a key bridge between traditional fossil fuels and renewable energy sources. We present a methodology for detecting anomalies in industrial natural gas consumption by transforming time series data into image representations using the Continuous Wavelet Transform (CWT). These scalograms, which preserve both temporal and frequency information, are used as inputs to convolutional neural networks (CNN) and a hybrid CNN–Long Short-Term Memory (LSTM) architecture designed to capture both spatial patterns and sequential dependencies. The models were trained and evaluated on real operational data from industrial customers, achieving F1-score above 0.80 in most cases. Exploratory analysis showed that normal and anomalous behaviors produce visually distinct wavelet-based patterns, supporting both supervised classification and human interpretability. Among the architectures tested, MobileNetV2 and the CNN-LSTM model offered the most stable and generalizable results. This work demonstrates that integrating wavelet transformations with deep learning enhances early and reliable anomaly detection in gas distribution systems.

Resumen

El sector del gas natural, desempeña un papel fundamental a nivel mundial en la actual transición energética, sirviendo de puente clave entre los combustibles fósiles tradicionales y las fuentes de energía renovables. Presentamos una metodología para detectar anomalías en el consumo de gas natural industrial mediante la transformación de datos de series temporales en representaciones de imágenes utilizando la Transformada de Wavelet Continua (CWT). Estos escalogramas, que conservan información temporal y de frecuencia, se utilizan como entradas para redes neuronales convolucionales (CNN) y una arquitectura híbrida CNN-Long Short-Term Memory (LSTM) diseñada para capturar patrones espaciales y dependencias secuenciales. Los modelos fueron entrenados y evaluados sobre datos operativos reales de clientes industriales, alcanzando puntuaciones F1 superiores a 0,80 en la mayoría de los casos. El análisis exploratorio mostró que los comportamientos normales y anómalos producen patrones basados en

wavelets visualmente distintos, apoyando tanto la clasificación supervisada como la interpretabilidad humana. Entre las arquitecturas probadas, MobileNetV2 y el modelo CNN-LSTM ofrecieron los resultados más estables y generalizables. Este trabajo demuestra que la integración de las transformaciones de wavelets con el aprendizaje profundo aumenta la detección temprana y fiable de anomalías en los sistemas de distribución de gas.

Keywords—Deep Learning, Neural Networks, Time Series Analysis, Wavelet Transform, Anomaly Detection, Natural Gas Consumption, Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM)

1. Introduction

In recent years, there has been a growing interest in optimizing the services provided by companies in various industries, particularly in the energy sector. The natural gas sector, in particular, plays a critical role worldwide in the current energy transition, serving as a key bridge between traditional fossil fuels and renewable energy sources [14]. Enhancing the monitoring and operational efficiency of gas distribution networks can provide significant benefits, including improved service reliability, increased safety, and reduced environmental impact [16].

However, the distribution of natural gas also presents considerable challenges. For distribution companies, early detection of anomalies, such as leaks, equipment malfunctions, or infrastructure deterioration, is essential to maintain operational efficiency and prevent serious environmental and economic consequences. For industrial consumers, undetected anomalies can lead to operational disruptions, financial losses, and potential regulatory penalties. Although distribution companies collect hourly data on gas pressure, temperature, and volume, the large volume and complexity of this information present significant difficulties for traditional monitoring systems, which often struggle to detect anomalous consumption patterns effectively. To address these challenges, energy companies are increasingly adopting intelligent systems and advanced data analytics to enhance the real-time monitoring and optimization of their distribution networks [16]. In this context, advanced

analytical techniques based on deep learning offer a promising solution.

This thesis explores the detection of anomalies in the gas consumption patterns of an industrial consumer through the use of multiscale signal analysis and deep learning. First, we apply the Wavelet Transform to the time series data, which helps break down the signal and highlight variations at different time scales. This process transforms complex data into images that reveal meaningful patterns. We then use a hybrid deep learning model that combines Convolutional Neural Networks (CNN) to extract visual features from these images and Long Short-Term Memory Networks (LSTM) to understand how those features evolve over time. This approach captures both the spatial and temporal characteristics of consumption behavior, improving the detection of subtle anomalies that can depend on timing (for example, a specific day of the week) or specific consumption patterns [2], [15]. Preliminary analyses using wavelet-based visualizations have revealed distinct consumption behaviors among certain industrial clients, suggesting that visual and temporal characteristics are valuable for anomaly detection. This enables the distribution company to identify atypical consumption patterns earlier, allowing for timely interventions, improved operational decision-making, and enhanced service reliability.

2. Related work

Anomaly detection in gas consumption systems has received growing attention as operational efficiency and reliability have become central concerns for distribution networks and utilities. Traditional methods focus on statistical and regression-based models. For instance, other works have proposed a two-stage probabilistic method to detect anomalies in natural gas time series using a linear regression model informed by weather variables [5], followed by a Bayesian maximum likelihood classifier. Similarly, others have developed a forecasting-based anomaly detection approach tailored to gas networks [8], using local regression to identify unusual demand behavior that could compromise network safety or operational efficiency.

On the other hand, the Wavelet Transform has been extensively applied to time series analysis because of its ability to represent local variations across several time scales. Our research demonstrated how wavelet transformations, including both orthonormal and bi-orthonormal variants, can outperform traditional transformations like the Fourier Transform in similar search tasks over high-dimensional time series [1]. In the same way, this research showed that wavelet-based representations maintain precision while enabling dimensionality reduction [1], which is essential in managing large volumes of operational data.

More recent applications have explored the integration of wavelets with deep learning techniques. Convolutional neural networks (CNN) trained on images transformed from wavelet-domain input to improve image classification tasks [3]. This approach effectively captured multiscale features and demonstrated superior accuracy compared to CNN trained on raw

pixel values. Similarly, other studies have shown that combining wavelet transformations with supervised learning models enhances the representation of temporal signals, making it more straightforward to identify clear patterns for recognition and classification.

In the context of deep learning for anomaly detection, CNN and recurrent architectures such as LSTM have emerged as powerful tools. Several authors have applied these models to network intrusion detection and time series classification problems. For example, significant improvement in intrusion detection has been achieved by combining CNN for spatial feature extraction and LSTM to capture temporal dependencies in benchmark datasets [9]. Hybrid models that fuse spatial and sequential features have been shown to reduce both false positives and false negatives, as evidenced by high precision and recall scores across multiple benchmark datasets. For example, using the CICIDS 2017 dataset, the SSA-LSTM model achieved a precision of 1.00 and a recall of 0.9444, with a testing accuracy of 0.998 [9]. Similarly, it reached a recall of 1.00 on the NSL-KDD dataset, further confirming its effectiveness in accurately identifying anomalies while minimizing false alarms.

Despite these advances, we were unable to find any work that directly combined wavelet-based visual representations with hybrid CNN-LSTM architectures for gas anomaly detection. Although individual components, such as wavelet transforms or CNN-LSTM models, have been applied separately to different domains, our work proposes a novel application that integrates these two components. Specifically, we propose an alternative method for anomaly detection in gas distribution nets that reframes the problem as an image classification task. The approach transforms gas consumption time series into wavelet-based scalograms, that is, visual representations of temporal signals, which are then processed by a CNN-LSTM model to capture both spatial energy patterns and temporal dynamics.

3. Data and Preprocessing

Before applying any deep learning technique, it is essential to ensure that the input data are properly structured, cleaned, and transformed. This section introduces the dataset used in this study and describes the steps taken to convert the time series into visual representations suitable for anomaly detection. The rationale behind variable selection, the criteria for labeling observations, and the procedures applied to standardize both image and numerical data for consistent and efficient modeling are shown in these section.

3.1 Dataset Description

The datasets used in this work contain operational data collected every hour from industrial customers connected to a natural gas distribution network. The study is based on two data sources:

- A **single-customer dataset**, containing detailed hourly consumption records for an industrial customer (101).
- A **multi-customer dataset**, which includes the same customer from the first dataset (101) along with two additional industrial customers (102 and 104), for a total of three.

Each dataset spans the period from **September 3, 2018** to **January 24, 2024**, covering over **five years** of uninterrupted activity and comprising exactly **46,849 hourly records** per customer. Every record is associated with a precise timestamp stored in the column `EFFECTIVE_DATE`. The datasets share the same schema and variable definitions; however, the consumption behaviors observed differ by customers, as will be described in subsequent sections.

The information was provided by the operating company under strict confidentiality agreements. Although the data corresponds to real operational behavior, client identifiers were anonymized. Each record includes the following variables: `RAW_VOLUME` and `STD_VOLUME`, representing the raw and pressure-temperature standardized gas volumes measured; `PRESSURE` and `TEMPERATURE`, which are physical measurements that directly influence volume correction calculations; and `ORIG_RAW_VOLUME`, `ORIG_STD_VOLUME`, `ORIG_PRESSURE`, and `ORIG_TEMPERATURE`, which are original values stored for traceability or auditing purposes.

The hourly resolution of the data enables the detection of short-term consumption patterns, such as daily cycles or operational shifts, while the long time span allows for the capture of seasonal variations and long-term behavioral trends. Preliminary exploration revealed that weekdays tend to exhibit higher and more stable consumption than weekends, which suggests that the gas demand is closely tied to the customer’s industrial operations, such as production schedules, equipment usage, or staffing routines, highlighting a consumption pattern driven by the operational calendar of the facility.

To better understand the structure and behavior of the datasets, Figure 1 and Figure 2 display the temporal evolution of all operational variables over the five-year period for each dataset separately. This visualization provides a first approximation for identifying relevant patterns, seasonality, and extreme deviations over time.

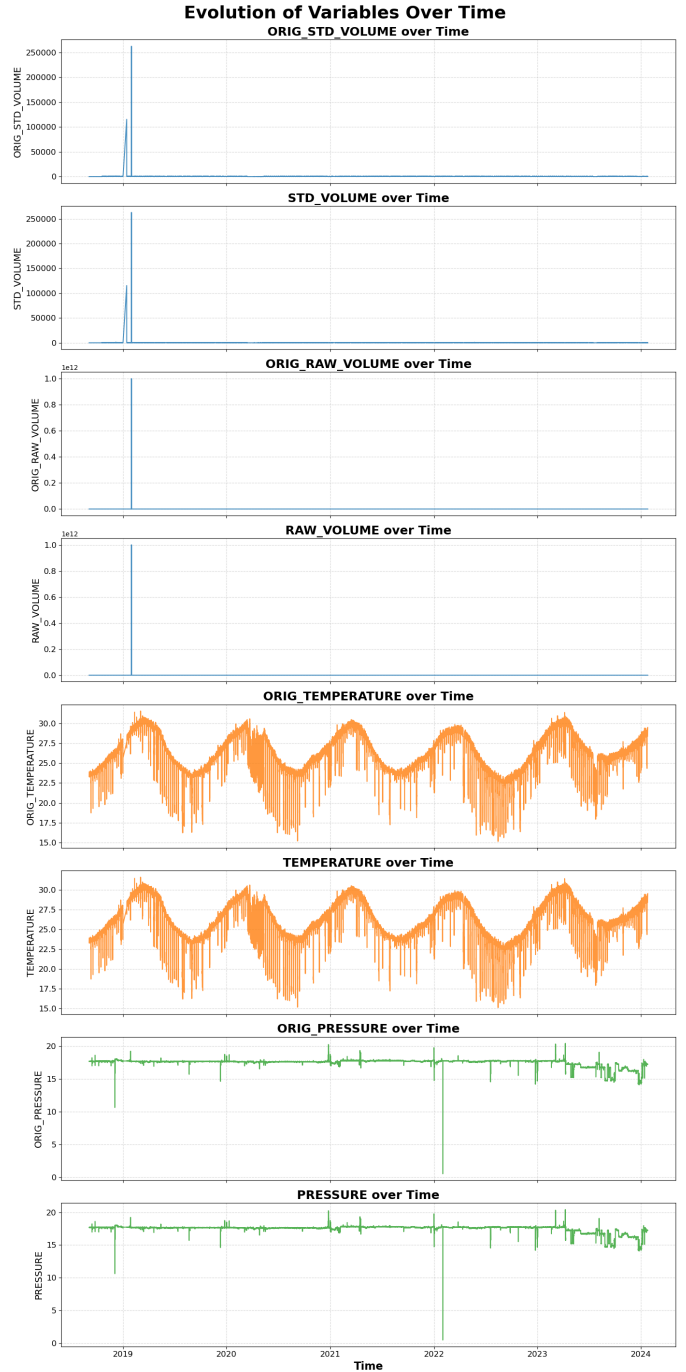


Fig. 1: Temporal evolution of operational variables related to Volume in the single-customer dataset.

In Figure 1, it is clear that temperature exhibits strong seasonal behavior, pressure remains mostly stable with small fluctuations, and volume variables—particularly `ORIG_RAW_VOLUME` and `STD_VOLUME`—show sharp spikes at specific moments in time. These abrupt changes, especially during the early years, suggest atypical operational conditions or measurement inconsistencies.

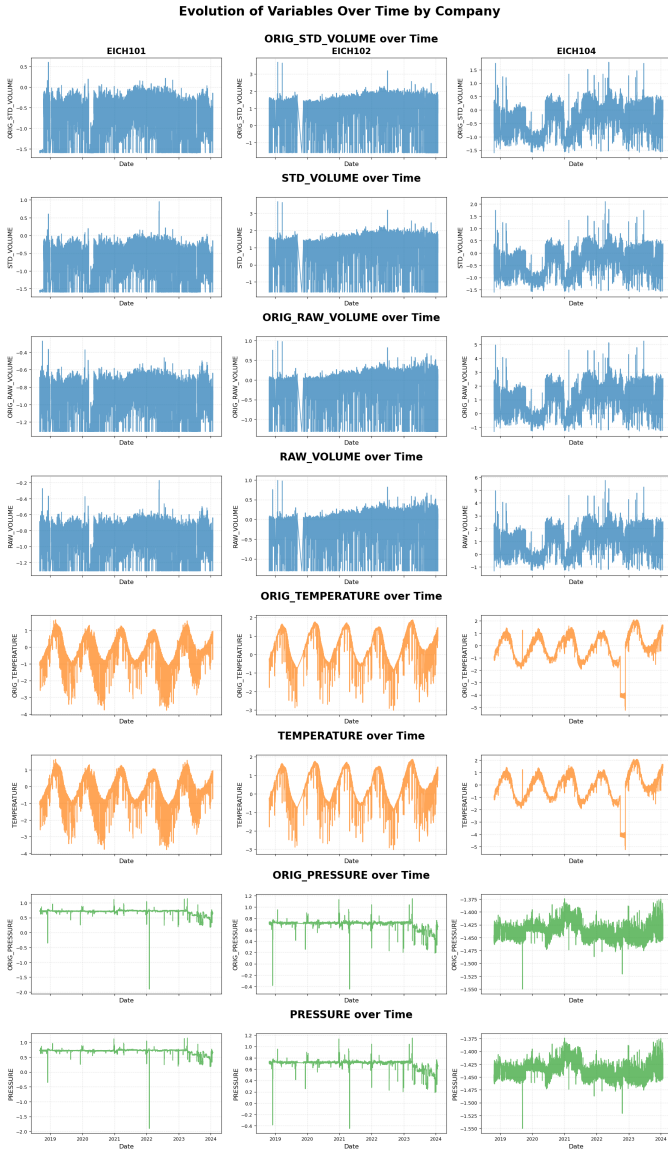


Fig. 2: Temporal evolution of operational variables related to Volume in the multi-customer dataset.

In Figure 2, it is evident that the three companies exhibit distinct patterns in their operational variables. Temperature, for all cases, demonstrates a clear seasonal cycle, reinforcing the influence of environmental conditions. Pressure remains generally stable over time, although minor fluctuations and abrupt drops suggest occasional disruption or sensor anomalies.

Customer 101 shows a relatively consistent behavior in volume-related variables, with minor variations and no significant outliers. This indicates steady operations with limited deviation. Customer 102, in contrast, shows a subtle upward trend in both standardized and raw volume values, potentially indicating a gradual increase in production or throughput over time. Customer 104, however, stands out because of its considerable variability in volume data. The presence of frequent peaks and irregular patterns suggests possible operational instability or more dynamic production profiles.

Given these distinct volume behaviors and their direct association with the companies' operational profiles, the volume-related variables emerged as the most informative signals for detecting anomalous patterns. Among all available variables, `ORIG_RAW_VOLUME` was selected as the main input to the models, as the focus of this study is on detecting anomalies in gas consumption patterns associated with the industrial client's operations. Unlike pressure or temperature, which are largely influenced by external environmental conditions, volume directly reflects the client's consumption behavior. Preliminary exploration showed that this variable exhibited a wide dynamic range and clear irregularities during anomalous events. While other variables showed high correlation and offered redundant information, `ORIG_RAW_VOLUME` provided a more direct and interpretable signal to identify unusual consumption patterns. For this reason, it was taken as the central variable of our work, with the possibility of adding contextual information, such as the date and hour associated with each measurement, for model enhancement.

To further illustrate the variability and presence of extreme values, Figure 3 and Figure 4 present a monthly box plot of the variable `ORIG_RAW_VOLUME`, showing how the distribution of gas consumption fluctuates over months and years. As shown in both figures, certain months exhibit a high concentration of outliers and wide interquartile ranges. However, not all extreme values necessarily represent anomalous behaviors; some may correspond to regular but variable operational behavior. This further supports the decision to use advanced signal processing techniques such as Wavelet Transform and machine learning models, which are better suited to distinguish between true anomalous behaviors and acceptable variability within the consumption pattern.

From Figure 3, monthly gas consumption shows considerable variability, with fluctuations in median values and dispersion across the period. Notably, months in 2020 and 2023 present narrower interquartile ranges and a higher frequency of outliers, suggesting irregular operational behavior. While no clear seasonal trend is observed, the presence of abrupt changes and scattered anomalies reinforces the relevance of this variable for detecting unusual consumption patterns.

From Figure 4, monthly gas consumption patterns differ notably between the three companies. 101 exhibits relatively stable distributions with limited dispersion and few outliers, indicating consistent usage. 102 presents a slight upward shift over time with broader interquartile ranges, suggesting gradual growth and more variable consumption. In contrast, 104 shows the highest volatility, with irregular medians, dispersed values, and frequent anomalies—pointing to unstable or highly dynamic operational behavior.

While the three companies display noticeable differences in their consumption behavior, Figure 4 reveals that they share a comparable overall range of variation. This overlap supports the decision to work with a combined dataset. Despite customer 101 showing more stable usage, customer 102 reflecting gradual growth, and customer 104 presenting more erratic patterns, the monthly distributions across the three clients

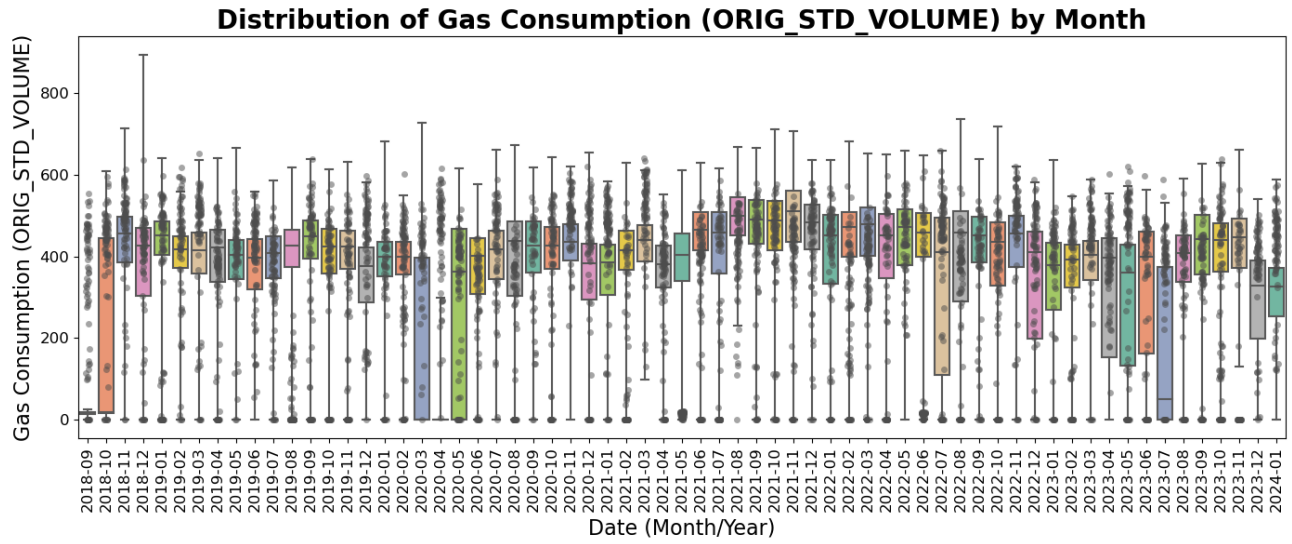


Fig. 3: Monthly distribution of gas consumption of the single-customer dataset (ORIG_RAW_VOLUME) from 2018 to 2024.

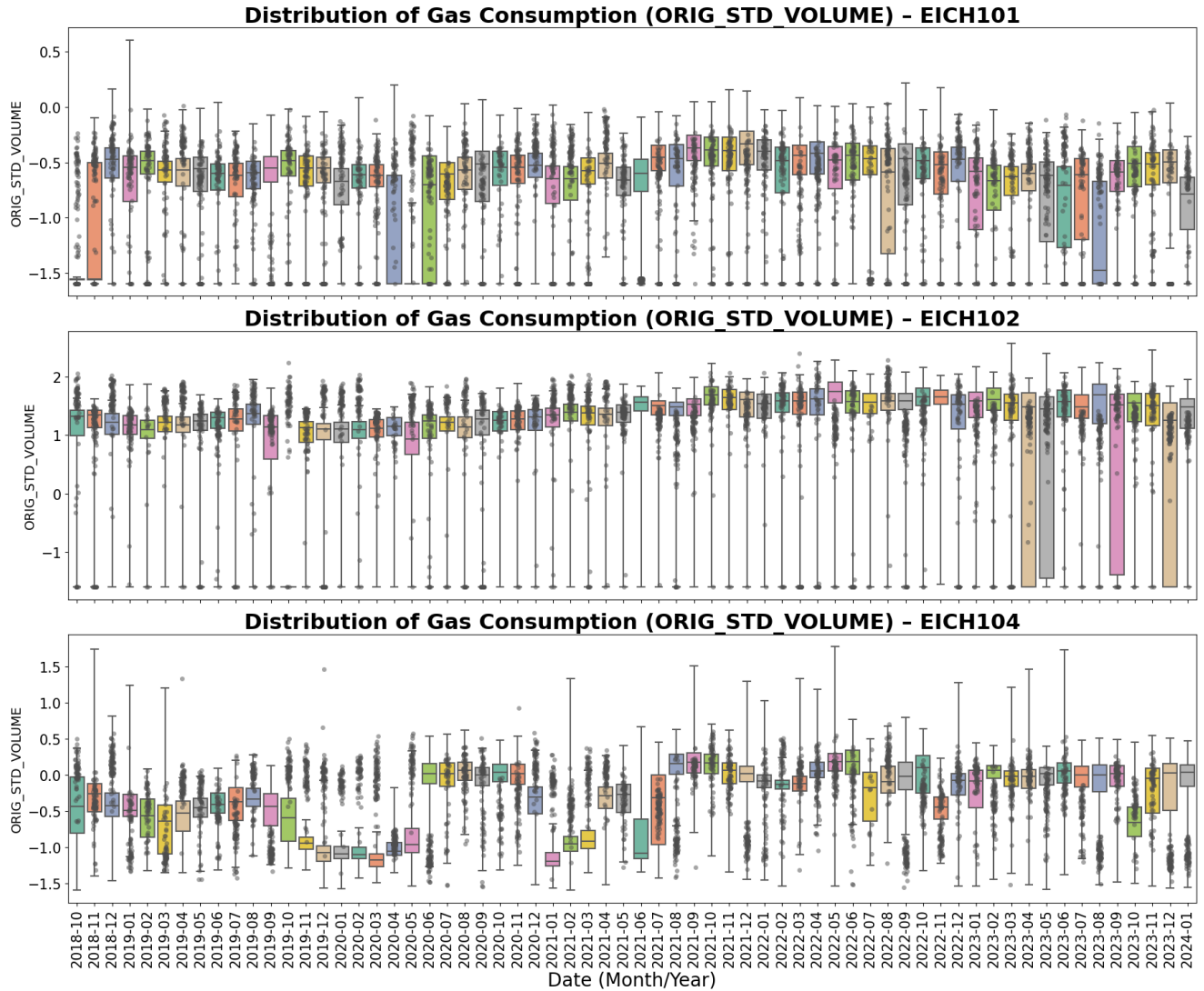


Fig. 4: Monthly distribution of gas consumption of the multi-customer dataset (ORIG_RAW_VOLUME) from 2018 to 2024.

remain within similar bounds in most periods. The presence of recurring outliers and fluctuations in each case suggests that, although their operational dynamics differ, they operate under a shared variability envelope. This common ground allows for the definition of a unified limit between typical and atypical behaviors, strengthening the basis for joint modeling and anomaly detection.

3.2 Image Generation

The transformation of gas consumption signals into image-based representations was motivated by the need to analyze temporal behaviors that are not easily detectable through raw data or classical time series techniques. Inspired by prior work in fault detection and signal classification, we explored a frequency-domain approach using the Continuous Wavelet Transform (CWT). This technique enables the generation of visual patterns that preserve both temporal dynamics and multiscale frequency components [6], [1].

Unlike the Fourier Transform, which provides only a global view of the frequencies present in a signal, the Continuous Wavelet Transform (CWT) decomposes signals in both time and scale, allowing for the detection of non-stationary patterns or localized anomalies that may not be evident otherwise. Mathematically, the CWT of a signal $x(t)$ is defined as:

$$CWT_x(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} x(t) \psi^* \left(\frac{t-b}{a} \right) dt \quad (1)$$

where a is the scale parameter, b is the translation parameter, and $\psi^*(t)$ is the complex conjugate of the mother wavelet function [17]. This formulation enables a localized analysis of signal content across different time windows and frequency bands.

Several works have leveraged this mathematical formula to improve performance in critical detection tasks. For example, it has been shown that applying the CWT provides a richer and more detailed time-frequency representation, which improves the detection of cyberattacks and electrical faults when used in conjunction with convolutional neural networks [6]. Similarly, wavelet-based representations derived from the CWT have proven effective in capturing local variations in frequency and shape while reducing data dimensionality, thereby facilitating visual comparison between signals and efficient reconstruction of behavioral patterns [1].

This representation was applied at two levels of temporal detail: weekly and daily. For the weekly case, CWT was used to process segments of seven consecutive days, while in the daily case, an image was generated per 24-hour period. Each signal was transformed into a scalogram and saved as a fixed-size RGB image (224x224), normalized to the $[0, 1]$ range. Figure 5 shows an example graph of the scalogram generated using the Wavelet Transform, which will serve as main input for model training and testing.

After generating all wavelet-based images, three independent datasets were created—one based on daily windows and weekly window of the single-customer dataset and another

weekly window of the multi-customer dataset. First, the daily set consists of **1,948 images** (970 labeled normal, 978 anomalous). Second, the weekly set of the single-customer includes **284 images** (157 labeled normal, 127 anomalous). These labeled image sets were then used for training and evaluating the respective CNN-based models. Third, the weekly set of the multi-customer includes **829 images** (449 labeled normal, 380 anomalous). These labeled image sets were then used for training and evaluating the respective CNN-based models.

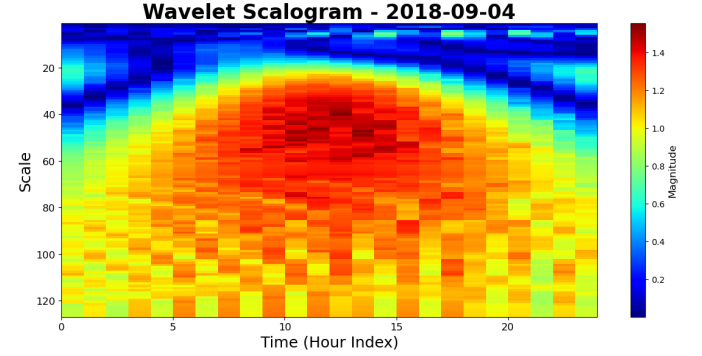


Fig. 5: Example of a scalogram generated per 24-hour period of the single-customer dataset

3.3 Exploratory Analysis and Pattern Discovery

Beyond their use as model inputs, the wavelet-based images revealed interpretable patterns that offered valuable insight into the customer’s operational behavior. These patterns varied depending on the temporal resolution and the dataset, and became a key element during the labeling process.

This preliminary exploration made it possible to identify those behaviors that predominate in the dataset, and those that are significant variations of the company’s behavior trend. These distinctions are not only useful for manual interpretation, but also make the problem suitable for classification of supervised images using convolutional neural networks, that you can learn to distinguish these visual signals with greater consistency and scalability than manual inspection.

3.3.1 Single-Customer Dataset Pattern Discovery

In the case of the single-customer dataset in the weekly images, a recurring shape resembling a “three-eyed” structure on the bottom of the image was frequently associated with regular consumption behavior; this pattern is illustrated in Figure 7. In contrast, daily representations of the single-customer dataset exhibited greater variability depending on the specific day; this pattern is illustrated in Figure 6. For example, Sundays typically showed two small energy clusters at the left-right extremes of the image, whereas Mondays displayed a warm-toned vertical band on the right and a circular concentration on the left. From Tuesday to Saturday, the weekly images tended to resemble the crescent-shaped pattern characteristic of the

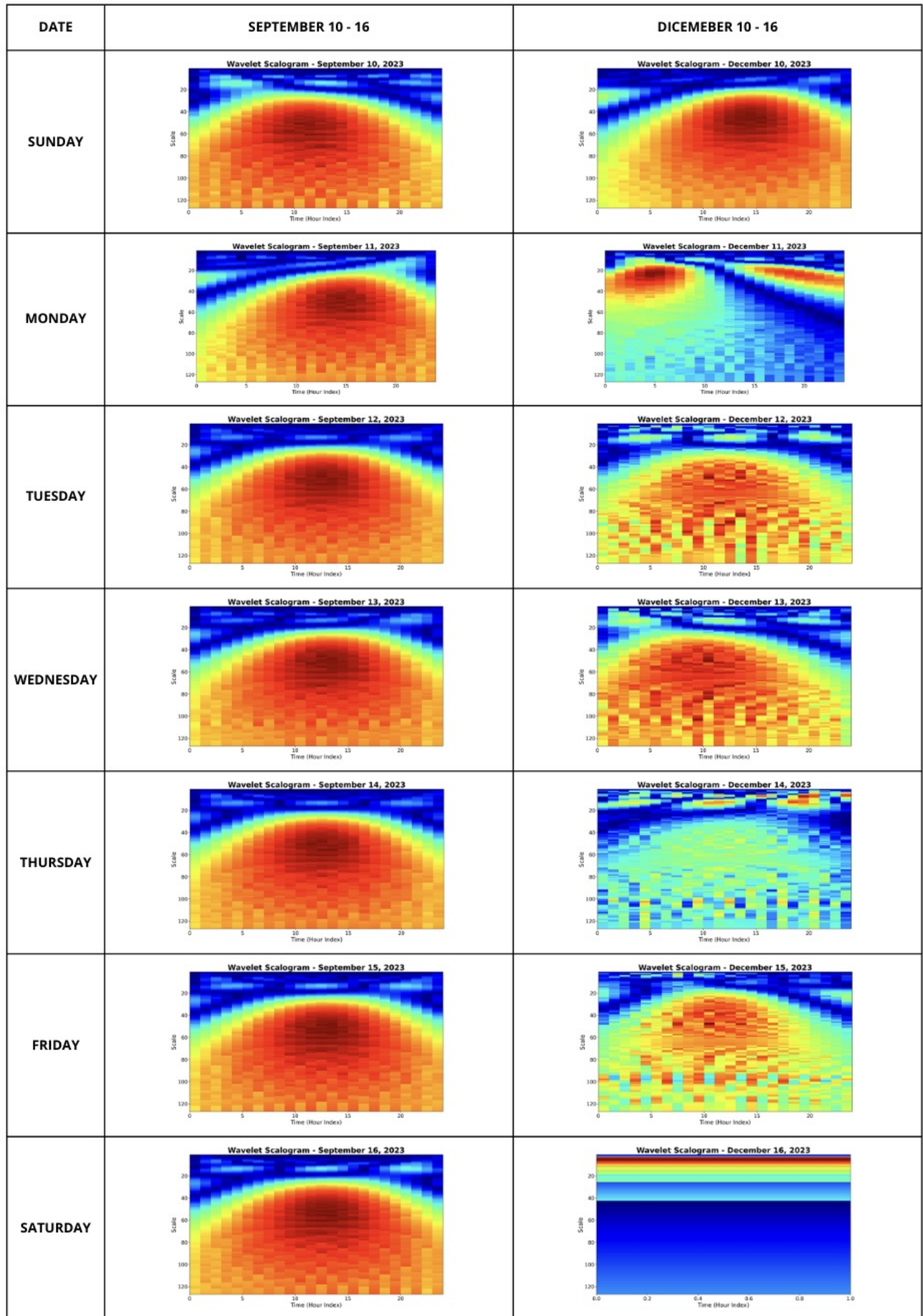


Fig. 6: Daily wavelet-based transformation of the single-customer dataset: (top) scalogram, (bottom) raw signal.

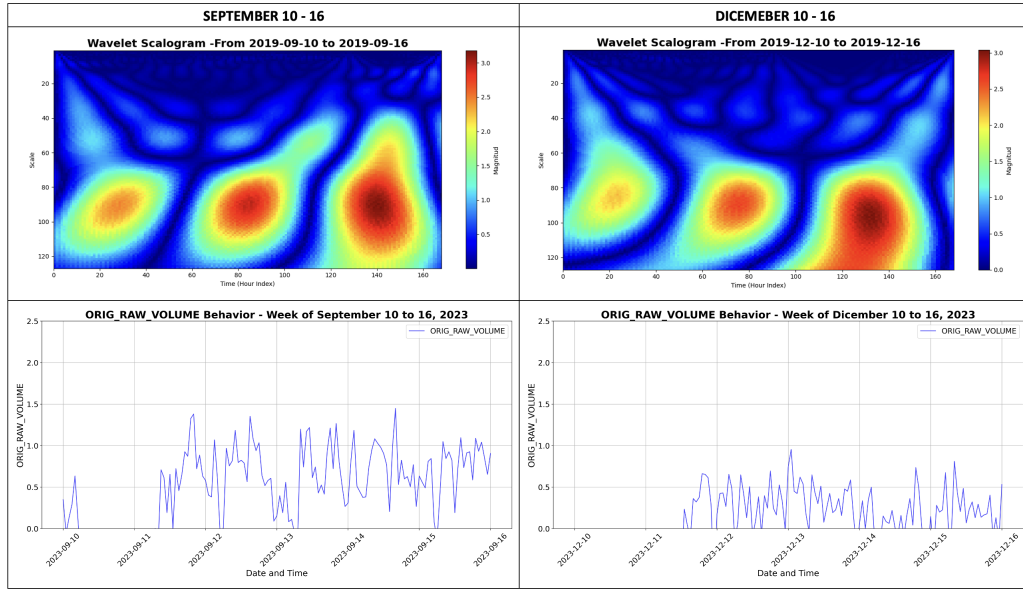


Fig. 7: Weekly wavelet-based transformation of the single-customer dataset: (top) scalogram, (bottom) raw signal.

daily signals. These recurring structures were key in guiding the manual annotation process and ensuring visual consistency across the labeled sets.

It is important to include the analysis of the raw signals. When comparing the behavior of the variable during the weeks of September 10-16 and December 10-16 of 2023 (bottom graphs in Figure 7), a significant difference in signal behavior is evident. In the first week (September), the variable shows a more pronounced and sustained variability, with relatively constant values oscillating around its average, which suggests regular and stable activity. In contrast, during the second week (December), the signal shows a noticeable decrease in its magnitude and frequency of oscillations, reflecting an atypical behavior. This observation is reinforced by the scale of the wavelet transform, where in September well-defined patterns of energy distributed along multiple scales are identified, while in December low energy is concentrated at high scales and limited time zones.

In contrast, some scalograms showed disruptions to these visual structures. Some showed an absence of expected high-energy regions, while others exhibited unexpected bursts or asymmetries across scales and time. These deviations were often subtle and not always visible in the raw time series, but became more apparent through their transformed representations. This observation supports the hypothesis that image-based representations using wavelets can enhance anomaly detection by highlighting features that are difficult to isolate through direct numerical analysis.

To further illustrate these irregularities, in Figure 6 it is possible to highlight the days that according to our analysis exhibit irregular behavior compared to the other images generated for the same day of the week. Tuesday, Wednesday, Thursday, Friday, and Saturday are the days that exhibit an irregular behavior because they display evident distortions or

a complete absence of structure (represented with the week of December 10-16). These anomalies include missing high-frequency clusters, asymmetric wavelet energy distributions, and flattened signal behavior, features that would be difficult to characterize without the combined time-frequency representation provided by the wavelet transform.

3.3.2 Multi-Customer Dataset Pattern Discovery

When analyzing the wavelet-based images of the multi-customer dataset, we observed a visual pattern that closely resembled the recurring “three-eyed” structure previously identified in the weekly images of the single-customer dataset. This pattern typically appears in the lower portion of the scalogram as a triad of warm-colored energy clusters, suggesting a similar underlying consumption behavior across different customers.

As in the single-customer case, several variations of this pattern were found. In some instances, the “three-eyed” structure appeared partially deformed or dissipated, transforming into irregular patches of blue-green tones that gradually disperse from the bottom to the upper regions of the image. Although this variation was not the dominant behavior within the dataset, it emerged consistently enough to be recognized as a recurrent anomaly class.

These visually consistent patterns provided valuable insights into the customer’s operational behaviors. The stability of certain shapes on specific days or time frames suggested the presence of regular cycles in gas usage, probably associated with their industrial activity. In particular, repetition of similar features for multiple weeks reinforced the idea that the signal exhibits strong temporal regularities, particularly under normal operating conditions.

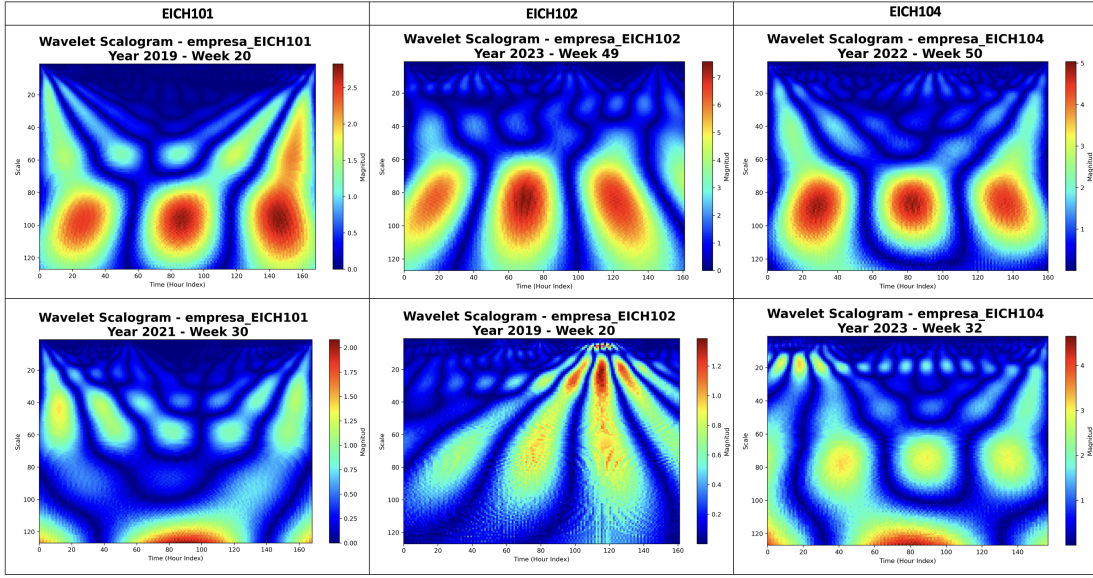


Fig. 8: Weekly wavelet-based transformation of the multi-customer dataset.

3.4 Image Labeling

The labeling process was performed manually to identify behavior patterns through visual analysis of raw signals and scalograms obtained by the CWT. Those days and weeks whose signals showed consistency in their patterns were categorized as normal, such as the stable and repetitive formations on the scalograms, characterized by energy concentrations at specific locations over time and frequency.

To start the labeling process, Figure 6 is a great tool because they illustrate the behavior during two different weeks of the year 2023, providing insight into how consumption patterns vary throughout the year. For example, those days that showed patterns consistent with energy distribution were considered normal, such as Mondays with vertical bands indicative of start activity and low-energy Sundays at the edges, reflecting a reduced transaction. In the case of weekly images from both datasets, weeks with repetitive patterns at the bottom of the scale, similar to a "three-eyed" formation, were labeled normal because they reflected stable operational cycles.

In contrast, those days and weeks whose signals showed significant variations from the expected patterns were classified as anomalous. These include the absence of typical energy structures, such as the disappearance of high frequency peaks, asymmetric energy distribution or the presence of unusual energy peaks. These irregularities were more evident in the scale charts than in the original signals, and reflected unusual behaviour such as variations in operational activity or system failures.

3.5 Preprocessing Steps

All scalograms generated from the wavelet transformation were saved in PNG format with a resolution of 224x224 pixels and three-color channels (RGB). Each image was encoded

with 8 bits per channel, resulting in a 24-bit color representation. This format preserved the visual detail of the frequency-time patterns while maintaining compatibility with standard convolutional neural network architectures.

The resulting data was represented as a tensor of shape (224, 224, 3). This preprocessing pipeline was applied uniformly to both the daily and weekly images of the single-customer dataset and the weekly images of the multi-customer dataset. It ensured that all model inputs shared the same format, dimensions, and scale, facilitating efficient training and reliable evaluation.

4. Methodology

The proposed methodology was validated using real operational data collected from an industrial gas distribution network. These data were used to evaluate whether applying the Continuous Wavelet Transform in combination with convolutional neural networks effectively improves anomaly detection performance

Two model architectures were developed and evaluated to address the task of gas consumption anomaly detection at different temporal resolutions. The proposed methodology explores two main approaches: (i) convolutional neural networks (CNN) applied to wavelet-transformed images for weekly pattern classification, and (ii) a hybrid CNN-LSTM architecture that jointly processes daily images and sequential numerical features. Each model is described in detail below, including their architectural design, input data structure, training strategy, and evaluation setup.

Performance was evaluated on the training and testing sets using accuracy, precision, and F1-score. The joint modeling of spatial and sequential signals aimed to reduce both false positives (Type I errors) and false negatives (Type II errors),

which are critical in anomaly detection scenarios. The following section outlines the metrics used to assess the model's performance in more detail.

- **Accuracy**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision**

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **F1-Score**

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

where:

$$\text{Recall} = \frac{TP}{TP + FN}$$

4.1 Weekly and Daily Anomaly Detection using CNN with Wavelet Transform Model Architecture

Two versions of the model were developed: one trained on a single-customer dataset to capture individualized consumption behavior, and another trained on a multi-customer dataset to evaluate generalization across diverse usage patterns. These models serve as a baseline for classifying weekly gas consumption patterns into anomalous or normal categories. Its performance provides a reference point for evaluating more complex architectures introduced later in this study. As previously described, the method uses a wavelet-based image representation of time-series data, which is then classified using a CNN. In this study, three widely used CNN architectures were evaluated as feature extractors in this study: **MobileNetV2**, **ResNet50**, and **VGG16**.

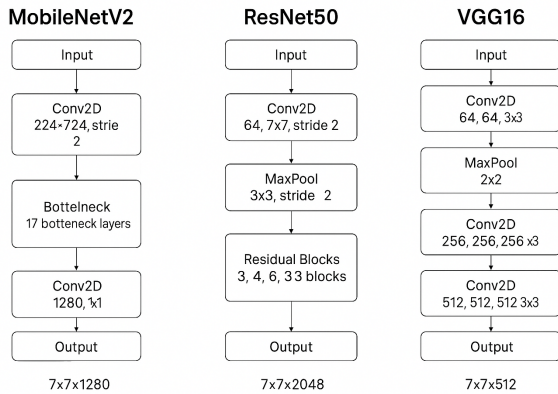


Fig. 9: Diagram comparing the architectural structure of MobileNetV2, ResNet50, and VGG16 as used for feature extraction in this study.

MobileNetV2, is a lightweight convolutional network designed for efficient inference in mobile and embedded applications. It uses depthwise separable convolutions and introduces inverted residual blocks with linear bottlenecks, significantly

reducing the number of parameters and operations while preserving accuracy [11].

ResNet50, addresses the vanishing gradient problem by implementing residual learning. It is composed of 50 layers organized in bottleneck blocks where shortcut connections allow the network to learn identity mappings, enabling the successful training of very deep architectures [12]

VGG16, is characterized by its simplicity: it stacks multiple 3×3 convolutional layers followed by max-pooling layers. Although relatively heavy in parameters, VGG16 established a strong baseline in several vision tasks due to its structured and deep design [13].

All three models were initialized with ImageNet weights and used as fixed feature extractors by freezing their convolutional bases. As detailed in Table I, each model was extended with a custom classification head consisting of a global average pooling layer (for MobileNetV2 and ResNet50) or a flattening layer (for VGG16), followed by a fully connected dense layer with 128 ReLU units and a final output neuron with sigmoid activation for binary prediction.

TABLE I: CNN Architectures and Classification Head Configuration

Base Model	Pooling Strategy	Classification Head
MobileNetV2	GlobalAveragePooling2D	Dense ((128, ReLU) + (1, Sigmoid))
ResNet50	GlobalAveragePooling2D	Dense((128, ReLU) + (1, Sigmoid))
VGG16	Flatten	Dense((128, ReLU) + (1, Sigmoid))

Models were trained using the Adam optimizer with a binary cross-entropy loss function for **10 epochs** and a **batch size of 32**. The dataset was randomly split into **70% training**, **20% validation**, and **10% testing** sets using a stratified split to maintain class balance.

4.2 Daily Anomaly Detection using CNN-LSTM Fusion Model Architecture

This model aims to capture fine-grained daily anomalies in gas consumption behavior by combining image-based and sequence-based data representations. The approach integrates two data modalities: (i) wavelet-transformed daily images, and (ii) numerical time series features aggregated per day. The rationale is that some anomalies may not be visually or statistically evident on their own, but emerge when both perspectives are jointly considered.

To effectively learn from these heterogeneous inputs, the outputs of the CNN (processing the image modality) and the LSTM (handling the sequential features) are concatenated and passed through a dense layer with 128 ReLU units, followed by a final sigmoid-activated output layer for binary classification. This fusion enables the model to make more informed decisions by leveraging both spatial and temporal patterns.

TABLE II: CNN-LSTM Architecture for Daily Anomaly Detection

Image Input	224x224 RGB, processed with MobileNetV2 (frozen)
Sequence Input	Daily features (8 variables), padded sequence
Fusion Strategy	Concatenation of CNN and LSTM branches
LSTM Layer	1 layer, 64 units
Dense Layer	Dense(128, ReLU) + Dropout(0.3) + Dense(1, Sigmoid)

a) Training Details.

The model consists of two parallel branches: a convolutional stream that processes RGB images and a recurrent stream that processes daily sequences of standardized operational variables. The convolutional stream uses a pre-trained **MobileNetV2** (frozen) as a feature extractor, followed by a global average pooling layer. The recurrent stream uses a single-layer **LSTM** with 64 units, fed by daily sequences of eight normalized variables: STD_VOLUME, ORIG_STD_VOLUME, TEMPERATURE, ORIG_TEMPERATURE, PRESSURE, ORIG_PRESSURE, RAW_VOLUME, and ORIG_RAW_VOLUME.

The model was trained with 20 epochs with the Adam optimizer, binary cross-entropy loss, and a batch size of 32.

5. Results

This section presents the evaluation results of the models developed for gas consumption anomaly detection. First, we analyze the performance of convolutional neural network (CNN) models trained on weekly and daily wavelet-transformed images, highlighting the differences among MobileNetV2, ResNet50, and VGG16 architectures. Then, we describe the results obtained with the CNN-LSTM hybrid model trained on daily data, assessing its ability to integrate visual and sequential information. Throughout this section, performance metrics such as accuracy, precision, recall, and F1-score are reported and discussed. Comparative insights are also provided to position the models' performance relative to each other and to existing literature.

5.1 Results of MobileNetV2 Single-customer Weekly Models

MobileNetV2 was trained using weekly scalograms derived from a single customer's gas consumption data. This setup aimed to capture individual behavioral patterns and detect anomalies with high sensitivity.

Figure 10 presents the training and validation loss and accuracy curves. The curves show a steady learning trajectory with limited signs of overfitting, suggesting effective convergence on a small and more homogeneous dataset.

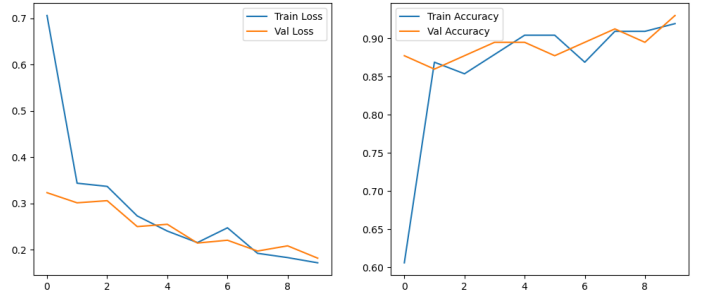


Fig. 10: Training and validation loss (left) and accuracy (right) for MobileNetV2 (single-customer weekly model)

The final performance metrics are summarized in Table III. The model reached an **F1-score of 0.93 on the training set, 0.93 on the validation set, and 0.97 on the test set**, indicating strong discriminative performance even in the more focused single-user scenario.

TABLE III: Performance of MobileNetV2 on Single-customer Weekly Anomaly Detection

Dataset	Accuracy	Precision	Recall	F1-score
Train	0.93	0.93	0.92	0.93
Validation	0.93	0.93	0.93	0.93
Test	0.97	0.97	0.96	0.97

To sum up, MobileNetV2 demonstrated reliable performance in detecting weekly anomalies for a single customer. The model maintained a balanced trade-off between false positives and false negatives across all dataset splits, confirming its robustness in highly personalized settings.

5.2 Results of ResNet50 Single-customer Weekly Models

The ResNet50 architecture was also evaluated in the single-customer weekly anomaly detection scenario. This approach allowed the model to specialize in individual consumption patterns.

Figure 11 displays the loss and accuracy curves. The training process showed gradual improvement in loss, with slightly more variation in accuracy. The validation performance was comparatively more stable, with no major divergence between metrics, though a gap between training and validation accuracy is noticeable, especially in the early epochs.

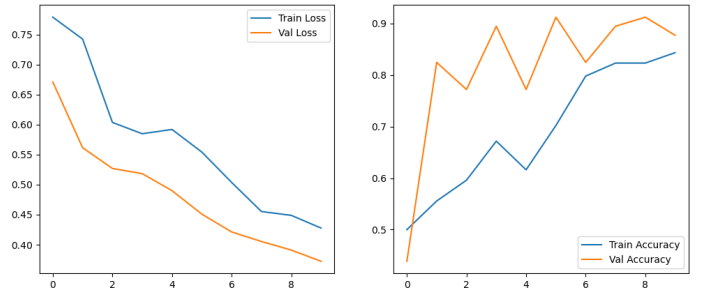


Fig. 11: Training and validation loss (left) and accuracy (right) for ResNet50 (single-customer weekly model)

The classification report for each dataset is summarized in Table IV. ResNet50 achieved an **F1-score of 0.82 on the training set, 0.87 on the validation set, and 0.89 on the test set**. While performance was competitive, the model exhibited a tendency toward false negatives, especially in the training partition.

TABLE IV: Performance of ResNet50 on Single-customer Weekly Anomaly Detection

Dataset	Accuracy	Precision	Recall	F1-score
Train	0.83	0.87	0.81	0.82
Validation	0.88	0.90	0.86	0.87
Test	0.90	0.91	0.88	0.89

In general, ResNet50 showed good capacity for detecting anomalies in a single-customer context. The larger number of false negatives in the training set suggests potential issues with recall that may benefit from further regularization or threshold tuning.

5.3 Results of VGG16 Single-customer Weekly Models

The VGG16 model training dynamics, shown in Figure 12, reveal a sharp initial drop in loss and a steady increase in accuracy, albeit with some fluctuations. The validation loss and accuracy remained relatively stable, indicating decent generalization capacity.

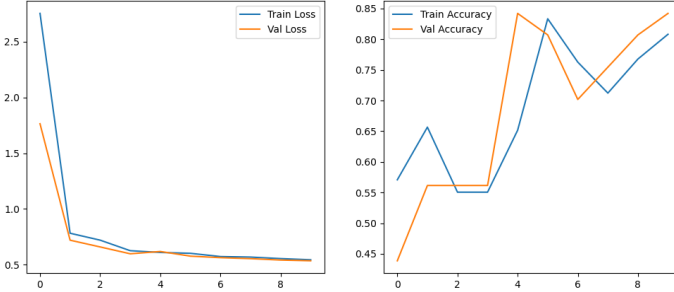


Fig. 12: Training and validation loss (left) and accuracy (right) for VGG16 (single-customer weekly model)

Performance metrics for the training, validation, and test sets are summarized in Table V. VGG16 achieved an **F1-score of 0.80 on the training set, 0.83 on the validation set, and 0.82 on the test set**. As with ResNet50, VGG16 showed a recurring pattern of false negatives across all datasets, especially in the training partition, where 37 anomalous cases went undetected.

TABLE V: Performance of VGG16 on Single-customer Weekly Anomaly Detection

Dataset	Accuracy	Precision	Recall	F1-score
Train	0.81	0.86	0.79	0.80
Validation	0.84	0.88	0.82	0.83
Test	0.83	0.87	0.81	0.82

On the whole, VGG16 achieved consistent results, although its tendency to miss anomalies (Type II errors) could limit its

applicability in more critical detection contexts. Nonetheless, it remains a competitive option with balanced performance across datasets.

5.4 Results of MobileNetV2 Multi-customer Weekly Models

To evaluate the weekly anomaly detection task, three CNN architectures were trained using scalograms generated through Continuous Wavelet Transform (CWT). The model **MobileNetV2**, demonstrated strong generalization capacity and stability across training, validation, and test sets.

Figure 13 shows the loss and accuracy curves during training. The validation metrics remained relatively stable, indicating that the model did not suffer from overfitting.

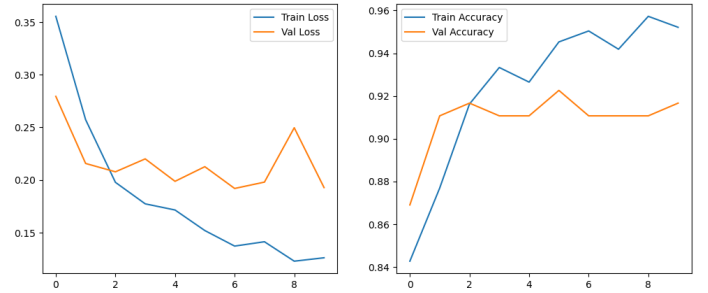


Fig. 13: Training and validation loss (left) and accuracy (right) for MobileNetV2

The final performance metrics are summarized in Table VI. The model achieved an **F1-score of 0.97 on the training set, 0.93 on the validation set, and 0.92 on the test set**, demonstrating strong discriminative capability.

TABLE VI: Performance of MobileNetV2 on Weekly Anomaly Detection

Dataset	Accuracy	Precision	Recall	F1-score
Train	0.97	0.97	0.96	0.97
Validation	0.93	0.93	0.92	0.93
Test	0.92	0.92	0.91	0.92

Altogether, MobileNetV2 demonstrated that visual representations of gas consumption using CWT scalograms are highly effective for anomaly classification. The results also show that the model was capable of generalizing to unseen data, which is a key requirement for real-world anomaly detection systems.

5.5 Results of ResNet50 Multi-customer Weekly Models

ResNet50 exhibited a more irregular training pattern compared to MobileNetV2. Figure 14 shows that both training and validation accuracy increased over epochs, although with some oscillations in the validation curve, suggesting higher sensitivity to batch variation.

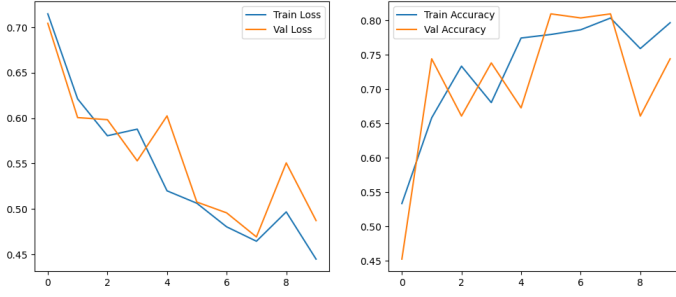


Fig. 14: Training and validation loss (left) and accuracy (right) for ResNet50

The model reached a test F1 score of **82%**, which is lower than MobileNetV2. Its recall and precision were also smaller compared to MobileNetV2, as summarized in Table VII.

TABLE VII: Performance of ResNet50 on Weekly Anomaly Detection

Dataset	Accuracy	Precision	Recall	F1-score
Train	0.88	0.91	0.81	0.86
Validation	0.89	0.91	0.83	0.87
Test	0.84	0.86	0.79	0.82

Globally, ResNet50 demonstrated reasonable classification performance across both training and testing sets. However, it exhibited higher variance in F1-scores, indicating less consistent performance compared to MobileNetV2. This increased variability suggests that the model is more sensitive to changes in the training data, which could hinder its reliability in production settings. Additionally, ResNet50 showed a greater tendency to generate false negatives, reducing its effectiveness in real-time anomaly detection where missing an anomaly can be critical. Although its generalization ability exceeded expectations—likely because deeper models like ResNet50 typically overfit on smaller datasets—its lower stability in precision and recall metrics compared to MobileNetV2 suggests that it may benefit from additional regularization techniques or architectural adjustments.

5.6 Results of VGG16 Multi-customer Weekly Models

The VGG16 model showed signs of overfitting during training. Although the model performed very well on the training set, the validation curve fluctuated, indicating that the model was fitting the training data too closely without generalizing as effectively. Figure 15 illustrates this behavior.

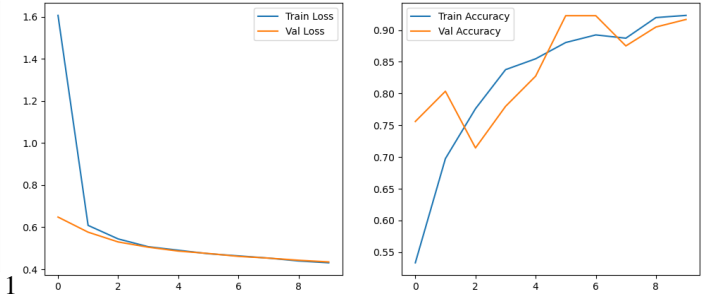


Fig. 15: Training and validation loss and accuracy for VGG16

Despite this, the model achieved competitive performance on the test set, with an **F1-score of 0.89**, indicating a balanced trade-off between precision and recall. Its nearly perfect training performance further supports the observation of overfitting.

TABLE VIII: Performance of VGG16 on Weekly Anomaly Detection

Dataset	Accuracy	Precision	Recall	F1-score
Train	0.99	1.00	1.00	1.00
Validation	0.94	0.96	0.91	0.93
Test	0.90	0.89	0.89	0.89

Overall, VGG16 achieved strong results in terms of accuracy and produced relatively balanced predictions across classes. However, its near-perfect performance on the training set, contrasted with noticeably lower scores on the validation and test sets, indicates overfitting. This suggests that the model has difficulty generalizing to unseen data. While its precision and recall remained competitive, its drop in performance outside the training set highlights lower robustness, which we assess in terms of generalization capability and consistency across different evaluation datasets. As such, VGG16 is a viable model, but a less robust alternative compared to MobileNetV2 and ResNet50, which maintained more consistent results across datasets.

5.7 Results of MobileNetV2 Daily Model

For the daily anomaly detection setting, MobileNetV2 was retrained using scalograms corresponding to individual days of gas consumption. Compared to the weekly setup, this task involved greater variability due to the increased resolution of input data—daily patterns can fluctuate significantly from one day to the next depending on consumer behavior, external conditions (e.g., weather), or operational anomalies. In contrast, weekly aggregates tend to smooth out such fluctuations, making the patterns more stable. Figure 9 shows the training and validation curves. While accuracy increased steadily, the validation loss showed more fluctuation, indicating moderate sensitivity to these day-to-day variations in the input data.

Figure 16 shows the training and validation curves. While accuracy increased steadily, the validation loss showed more fluctuation, indicating moderate sensitivity to daily variations.

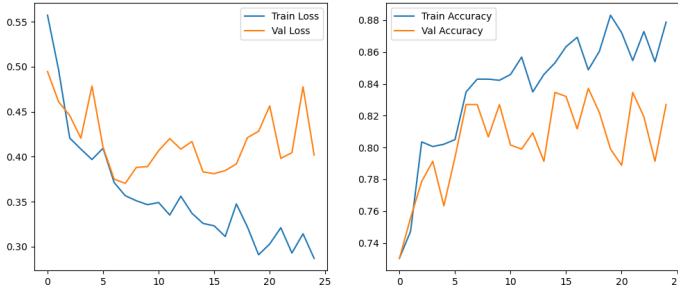


Fig. 16: Training and validation loss (left) and accuracy (right) for MobileNetV2 - Daily

As summarized in Table IX, the model achieved an **F1-score of 0.90** on the training set, **0.83** on the validation set, and **0.74** on the test set. These results reflect good performance given the complexity of daily consumption patterns, although the decrease in generalization, compared to the training performance, indicates that further tuning or regularization may be needed.

TABLE IX: Performance of MobileNetV2 on Daily Anomaly Detection

Dataset	Accuracy	Precision	Recall	F1-score
Train	0.90	0.90	0.90	0.90
Validation	0.83	0.83	0.82	0.83
Test	0.74	0.74	0.74	0.74

In summary, MobileNetV2 showed robustness under the daily classification setting, which we define as the model's ability to maintain stable performance across training, validation, and test sets despite increased input variability. Although there was a noticeable drop in generalization performance compared to the weekly version—as seen in the higher gap between training and validation metrics—the model still effectively captured short-term anomalies and maintained a reasonable balance between false positives and false negatives.

5.8 Results of ResNet50 Daily Model

ResNet50 was also evaluated for daily anomaly detection using scalograms as input. As seen in Figure 17, both training and validation curves showed noisy behavior, with several fluctuations throughout the epochs. This suggests instability in learning, likely due to the higher complexity and variability of daily patterns.

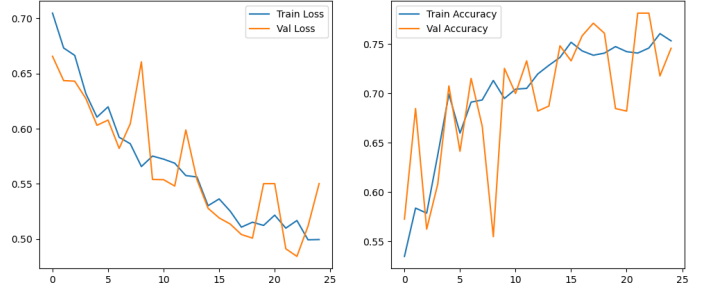


Fig. 17: Training and validation loss (left) and accuracy (right) for ResNet50 - Daily

Table X summarizes the performance across datasets. The model obtained an F1-score of **0.71** on the training set, **0.73** on validation, and **0.70** on the test set. While precision remained high across splits, recall was unbalanced, with a clear bias toward detecting anomalous days (class 1), often at the cost of false positives.

TABLE X: Performance of ResNet50 on Daily Anomaly Detection

Dataset	Accuracy	Precision	Recall	F1-score
Train	0.73	0.80	0.73	0.71
Validation	0.75	0.80	0.75	0.73
Test	0.72	0.79	0.72	0.70

Overall, while ResNet50 showed acceptable performance for daily detection, its imbalance in class prediction and unstable learning curve suggest that further regularization or data augmentation may be required to improve its reliability in production contexts.

5.9 Results of VGG16 Daily Model

The VGG16 model showed the most stable and progressive learning behavior among the evaluated CNN for daily detection. As illustrated in Figure 18, both training and validation curves suggest effective convergence with moderate overfitting. Despite this, the general trends show that the model was still able to learn meaningful patterns from the data.

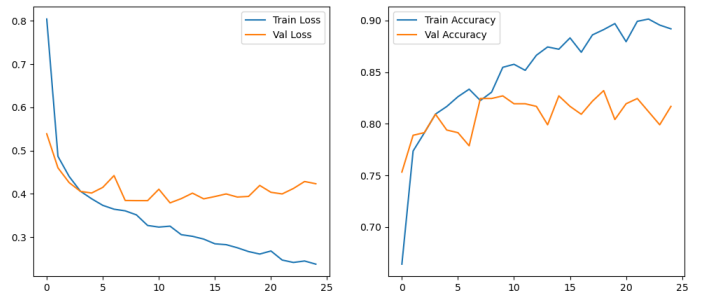


Fig. 18: Training and validation loss (left) and accuracy (right) for VGG16 - Daily

Performance metrics are summarized in Table XI. The model achieved an F1-score of **0.91** on the training set, **0.82** on the validation set, and **0.76** on the test set. These values

indicate that VGG16 generalized reasonably well, maintaining good balance between precision and recall.

TABLE XI: Performance of VGG16 on Daily Anomaly Detection

Dataset	Accuracy	Precision	Recall	F1-score
Train	0.91	0.91	0.91	0.91
Validation	0.82	0.83	0.82	0.82
Test	0.76	0.76	0.76	0.76

In general, VGG16 proved to be a strong candidate for daily anomaly detection, achieving a better balance between stability and generalization than ResNet50. It also offered more consistent results in the test partition, meaning that its performance metrics—particularly accuracy, precision, and recall—varied less across multiple training runs and were closer to its validation performance, suggesting lower sensitivity to initialization or random fluctuations in the data.

5.10 Results of MobileNetV2+LSTM Daily Model

To evaluate the benefit of combining visual and sequential information, we implemented a hybrid model that fuses CNN-based features extracted from daily scalograms with numerical time-series features processed by an LSTM layer.

Figure 19 shows the training dynamics. Both the loss and accuracy curves indicate stable learning without significant overfitting, although the validation performance fluctuated more than in the weekly experiments.

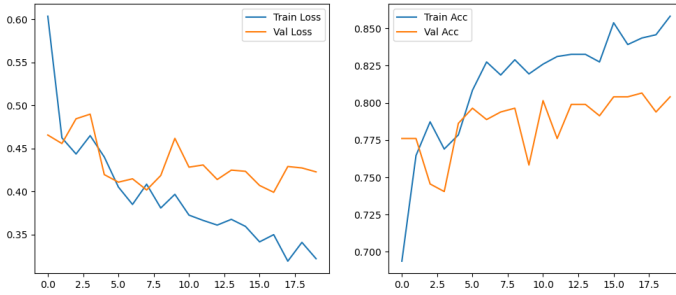


Fig. 19: Training and validation loss (left) and accuracy (right) for CNN+LSTM daily model

The model achieved competitive results across all partitions, with an **F1-score of 0.88 on training, 0.81 on validation, and 0.81 on test**. Table XII summarizes its final performance.

TABLE XII: Performance of CNN+LSTM on Daily Anomaly Detection

Dataset	Accuracy	Precision	Recall	F1-score
Train	0.88	0.88	0.88	0.88
Validation	0.81	0.82	0.80	0.81
Test	0.81	0.82	0.81	0.81

In sum, the CNN+LSTM model achieved the most balanced results across all datasets, meaning it maintained a good trade-off between precision and recall, with relatively high F1-scores and low variability in performance across training, validation,

and test sets. This confirms the advantage of combining temporal signals with visual representations of gas consumption patterns, as it allowed the model to capture both sequential and structural aspects of the data more effectively than single-modality models.

6. Discussion

This research proposed a methodology for detecting anomalies in industrial gas consumption by using multiscale signal transformations and deep learning architectures. Through the application of the Continuous Wavelet Transform (CWT), raw time series data were converted into scalograms that preserved both temporal and frequency-domain information. These images were then analyzed using convolutional neural networks (CNN) and a hybrid CNN–Long Short-Term Memory (LSTM) model to capture spatial patterns and sequential dependencies, respectively.

The visual nature of the wavelet-based representations proved essential not only for model training but also for exploratory analysis and manual labeling. Recurrent structures in normal consumption behavior, such as the crescent or “three-eyed” patterns, became visually recognizable and interpretable across both daily and weekly resolutions. In contrast, anomalous behaviors manifested as distortions, asymmetries, or the absence of expected energy clusters. These patterns would have been difficult to identify through traditional time-series techniques, reinforcing the value of transforming consumption data into image-based formats.

In terms of model performance, MobileNetV2 consistently delivered the best balance between generalization and accuracy across both weekly and daily tasks. The hybrid CNN-LSTM model, which incorporated numerical time-series features alongside image inputs, achieved the most stable performance in the daily detection scenario, confirming the added value of integrating multimodal data. While VGG16 and ResNet50 also showed competitive results, they were more prone to overfitting or variability in recall, making them less robust in practice. The general performance of the proposed models is summarized in Table XIII. The resolution was defined as:

- **Weekly (Multi)** → WM
- **Weekly (Single)** → WS
- **Daily (Multi)** → DM

TABLE XIII: Summary of model performance across all experiments

Model	Resolution	Accuracy	Precision	Recall	F1
MobileNetV2	WM	0.92	0.92	0.91	0.92
ResNet50	WM	0.84	0.86	0.79	0.82
VGG16	WM	0.90	0.89	0.89	0.89
MobileNetV2	WS	0.97	0.97	0.96	0.97
ResNet50	WS	0.90	0.91	0.88	0.89
VGG16	WS	0.83	0.87	0.81	0.82
MobileNetV2	DS	0.74	0.74	0.74	0.74
ResNet50	DS	0.72	0.79	0.72	0.70
VGG16	DS	0.76	0.76	0.76	0.76
MobileNetV2+LSTM	DS	0.81	0.82	0.81	0.81

Compared to related literature, the proposed models demonstrate competitive and, in many cases, superior performance. For instance, Akouemo and Povinelli (2016) [5] reported probabilistic anomaly detection using regression-based methods with performance metrics of 0.87 in precision, constrained by linear assumptions and exogenous variables. Similarly, Baldacci et al. (2016) [8] achieved effective results using local regression for anomaly detection in gas demand but lacked multiscale temporal representation.

The use of wavelet transforms, as supported by Popivanov and Miller (2002) [1], significantly improves signal interpretability—yet these were not previously combined with deep learning for gas anomaly detection. Our hybrid CNN-LSTM model, integrating CWT-based images with sequential sensor data, consistently surpassed the 0.80 F1-score benchmark on real-world data, confirming the benefits of multimodal representation. These findings suggest that combining wavelet-based visual input with deep neural architectures is a promising direction for robust and interpretable anomaly detection in the energy sector.

From a practical point of view, the methodology proposed in this thesis offers promising results for operational deployment. The high classification scores achieved in both temporal resolutions suggest that this approach could be used to automate anomaly detection in gas distribution systems, enabling earlier interventions and more efficient resource allocation. Additionally, the interpretability of the wavelet-generated scalograms enables a collaborative diagnostic process, in which automated classifications can be complemented by expert visual analysis.

7. Conclusion

Despite these strengths, several limitations must be acknowledged. First, the labeling process was based on visual inspection, which allowed a consistent initial annotation of the data. Although this approach proved useful for exploratory purposes, future work could enhance it further by incorporating expert feedback or adopting semi-supervised learning techniques to strengthen the robustness and generalizability of the labeling process. Second, the study was conducted using data from a single industrial client, which may limit the generalizability of the findings. Applying this methodology to diverse clients and operational contexts would be essential to evaluate its broader applicability.

Future research may also consider extending this framework by experimenting with attention mechanisms or transformer-based architectures, which have shown promising results in time-series anomaly detection. Furthermore, the exploration of other time-frequency representations, such as short-time Fourier transforms or empirical mode decompositions, could provide alternative or complementary insights to those derived from CWT.

In conclusion, this work contributes a scalable, interpretable, and accurate approach to industrial gas anomaly detection. By combining the strengths of wavelet signal processing and deep learning, it advances the capabilities of data-driven monitoring

systems in energy distribution networks. The integration of spatial and temporal representations into a unified classification framework represents a meaningful step toward smarter and more resilient infrastructure management.

References

- [1] Popivanov, I., & Miller, R. J. (2002). Similarity search over time-series data using wavelets. *Proceedings of the 18th International Conference on Data Engineering (ICDE)*, 233–243. <https://doi.org/10.1109/ICDE.2002.994711>
- [2] Zhao, X. (2022). *Wavelet-Attention CNN for Image Classification*. arXiv preprint arXiv:2201.09271. <https://arxiv.org/abs/2201.09271>
- [3] Wang, L., & Sun, Y. (2022). *Image classification using convolutional neural network with wavelet domain inputs*. *IET Image Processing*, 16(8), 2037–2048. <https://doi.org/10.1049/ipr2.12466>
- [4] Aditi, M. K. N., & Poovammal, E. (2019). *Image classification using a hybrid LSTM-CNN deep neural network*. *International Journal of Engineering and Advanced Technology (IJEAT)*, 8(6), 1342–1348. <https://doi.org/10.35940/ijeat.F8602.088619>
- [5] Akouemo, H. N., & Povinelli, R. J. (2016). *Probabilistic anomaly detection in natural gas time series data*. *International Journal of Forecasting*, 32(3), 948–956. <https://doi.org/10.1016/j.ijforecast.2015.06.001>
- [6] A. Abu Nassar, W.G. Morsi. (2024). *Detection of Cyber-Attacks and Power Disturbances in Smart Digital Substations using Continuous Wavelet Transform and Convolution Neural Networks*. *IEEE Access*, 9, 108784–108796. <https://doi.org/10.1016/j.j.epr.2024.110157>
- [7] Alhussein, M., Alharbi, A., Aydin, N., & Alshazly, H. (2023). *Detection of Cyber Attacks on Smart Grids Using Improved VGG19 Deep Neural Network Architecture and Aquila Optimizer Algorithm*. *IEEE Access*, 11, 88342–88356. <http://dx.doi.org/10.21203/rs.3.rs-3217829/v1>
- [8] Baldacci, L., Golfarelli, M., Lombardi, D., & Sami, F. (2016). *Natural gas consumption forecasting for anomaly detection*. *Expert Systems with Applications*, 62, 190–201. <https://doi.org/10.1016/j.eswa.2016.06.013>
- [9] Dash, N., Chakravarty, S., Rath, A. K., & Tripathy, R. (2025). *An optimized LSTM-based deep learning model for anomaly network intrusion detection*. *Scientific Reports*, 15, 1554. https://www.researchgate.net/publication/387897520_An_optimized_LSTM-based_deep_learning_model_for_anomaly_network_intrusion_detection
- [10] P. Wu and H. Guo. (2019). *LuNet: A Deep Neural Network for Network Intrusion Detection*. *Applied Soft Computing*, 113, 107924. <https://ieeexplore.ieee.org/abstract/document/9003126>
- [11] Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L. C. (2018). *MobileNetV2: Inverted Residuals and Linear Bottlenecks*. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 4510–4520. doi: 10.1109/CVPR.2018.00474.
- [12] He, K., Zhang, X., Ren, S., & Sun, J. (2016). *Deep Residual Learning for Image Recognition*. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778. doi: 10.1109/CVPR.2016.90.
- [13] Simonyan, K., & Zisserman, A. (2014). *Very Deep Convolutional Networks for Large-Scale Image Recognition*. arXiv preprint arXiv:1409.1556. <https://arxiv.org/abs/1409.1556>
- [14] International Energy Agency (IEA) (2024), *The Role of Gas in Today's Energy Transitions*, IEA Report, 2019. Available at: <https://www.iea.org/reports/the-role-of-gas-in-todays-energy-transitions>
- [15] Kanarachos, S., Christopoulos, S.-R., Chroneos, A., et al., *Detecting anomalies in time series data via a deep learning algorithm combining wavelets, neural networks and Hilbert transform*, *Expert Systems with Applications*. <https://doi.org/10.1016/j.eswa.2017.04.028>

- [16] Industrial Scientific. (2022) *Understanding Real-Time Area Gas Monitoring: Importance and Benefits*, Connected Safety—Industrial Scientific <https://www.indsci.com/en/blog/understanding-real-time-area-gas-monitoring-importance-benefits>
- [17] Torrence, C., & Compo, G. P. (1998). *A practical guide to wavelet analysis*. Bulletin of the American Meteorological Society, 79(1), 61–78. [https://doi.org/10.1175/1520-0477\(1998\)079%3C0061:APGTWA%3E2.0.CO;2](https://doi.org/10.1175/1520-0477(1998)079%3C0061:APGTWA%3E2.0.CO;2)